Multi-Agent Systems (BSc Artificial Intelligence, XB_0024)

Koen Hindriks, Mike Ligthart, Alexia Muresan

Assignment 0

Published: 28 October 2025

Submission date: 3 November 2025 (via Canvas, before 23:59)

Name, student ID: ....

Grade: .... [Pass/Fail], need a pass in order to follow the rest of the course! Please notify your TA when despite your efforts you keep running into issues.

Introduction

Cognitive agents introduce a novel computing paradigm for software application development and agent-based simulations that can be used to model interactions and emergent properties. In the Multi-Agent Systems (MAS) course, you will learn how to create and develop software environments that consist of multiple interacting cognitive agents. More specifically, you will be taught how to develop a MAS in which the agents use knowledge representation for reasoning and an agent programming language for decision-making in a simulated environment.

The objective of this assignment is to prepare and install all the tools that you will need for completing the assignments in the course. The course consists of two parts. The first part introduces the language for reasoning that we will use, and the second part introduces the agent language for decision-making. For the reasoning part, we will use the [Prolog programming language]. For the decision-making part, we will use the [MARBEL agent programming language].

This assignment does not carry any marks, but you will get a status of pass or fail. You need to work individually on this assignment. Support from Teaching Assistants (TAs) is available if necessary.

Deliverables

Submit a PDF file on Canvas with the status (Pass/Fail) of completing each of the five tasks below.[1]

[1] If computer-specific dependencies prevent you from completing a task, and you get an error, then you need to explain the specific problem you encountered with supportive data (e.g., screenshots, error log, etc.) in the PDF you submit. Make sure you have contacted TA for support to fix any issues you encounter.

Assignment

You need to perform the following tasks to complete the assignment:

Task-1: Installing and executing SWI Prolog

The purpose of this task is to get SWI Prolog, a specific implementation of Prolog, up and running.

1. *Download SWI Prolog* for your operating system from here.
2. *Install SWI Prolog*. Run the installer and follow the instructions.
3. *Launch (start) SWI Prolog*.
4. *Create a folder on your file system* where you want to save the Prolog files (`*.pl`) that you create, e.g.,

   `C:/Prolog/assignment-0`

5. *Change the working directory of your SWI Prolog system* with the command below:

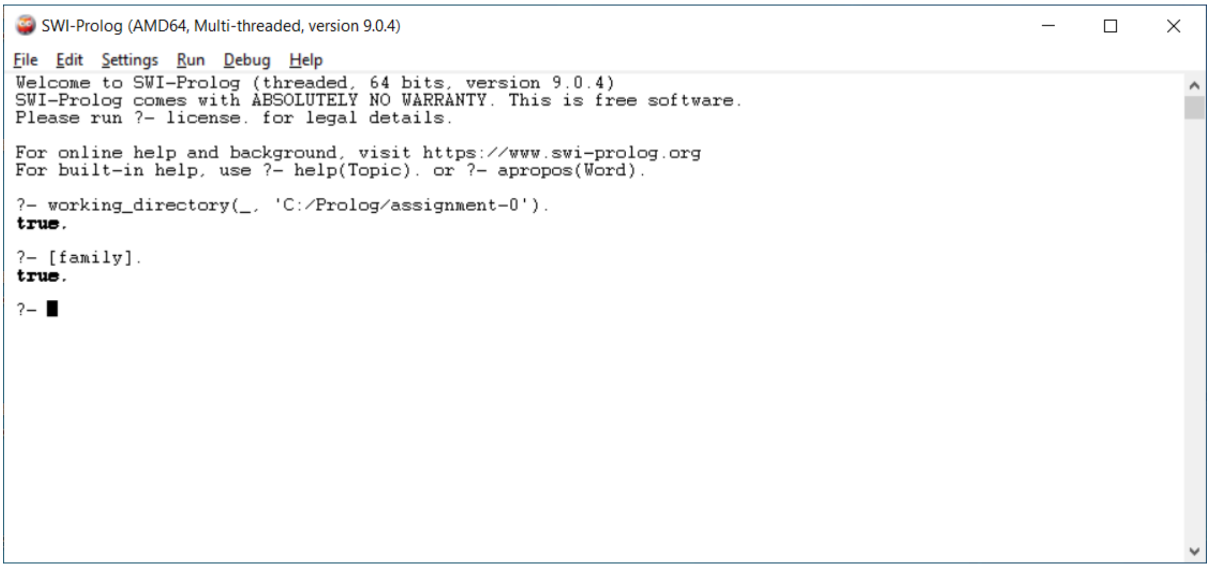
   `working_directory(_,'C:/Users/obama/PrologAssignments').`

   (replace the folder name in this example with the name of the folder you created in step 4.) Make sure to keep the quotes (') and to use slashes rather than backslashes!

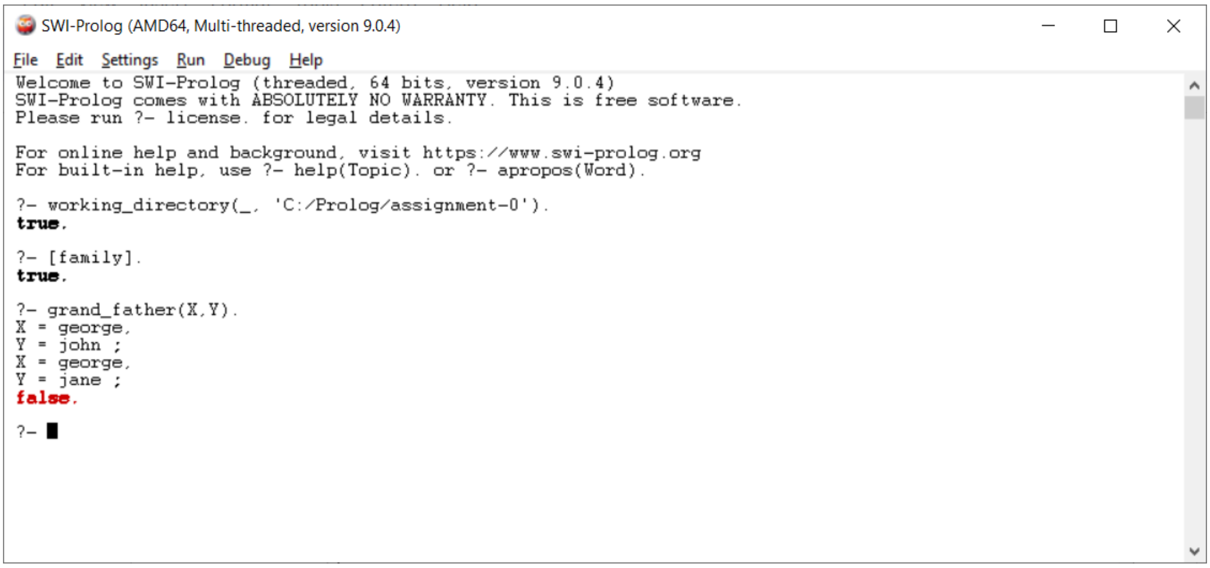
6. *Create a new Prolog file*. Using the *File>New* menu, create a file that you call `family.pl` and copy the code below into the editor that opens. Save the file (*File>Safe buffer*).

```
% Prolog database for the family example.

child(john,sue).   child(john,sam).
child(jane,sue).   child(jane,sam).
child(sue,george). child(sue,gina).

male(john).    male(sam).     male(george).
female(sue).   female(jane).  female(june).

parent(Y,X)  :- child(X,Y).
father(Y,X)  :- child(X,Y), male(Y).

grand_father(X,Z)  :- father(X,Y), parent(Y,Z).
```

7. *Consult the Prolog code* you just saved by reading the file by executing `[family].` (don't forget the dot!) in your SWI Prolog console. You should see the following output:



8. In the console, query who is grandparent of whom using the `grand_father/2` predicate. To get all the answers, just type semicolon as shown in the image below. If you see the output below, then you have successfully installed Prolog.



9. *Terminate the console* by executing the SWI Prolog halt command. Type in `halt.` (remember to also add the period at the end!) and then hit the ENTER key.

Task-2: Installing Java and Eclipse

The purpose of this task is to get Eclipse up and running with Java 11. If you already have Eclipse (latest version) and Java 11 installed on your system, then you can directly move to step 6:

1. *Download Java 11*:

    a. NOTE: If you have already installed (another) version of Java, *we advise you to first make sure to uninstall that version*. Although it is possible to use multiple Java versions on your system, it may complicate things and your system may not be able to locate the Java 11 version that we need for this course.

b. Go to https://adoptium.net/temurin/releases/?version=11 or to https://www.oracle.com/java/technologies/downloads/#java11[2].

c. Select your operating system and download an installer.

2. *Install JDK-11*[3] using the installer you downloaded and validate whether it is fully working. You can find guided instructions on how to validate this here. Output in your terminal window should show something similar to (amongst other things):

   Java(TM) SE Runtime Environment 18.9 (build 11.0.28+9-LTS-256).

3. *Download* the most recent version of *Eclipse* for your operating system here.

4. *Install Eclipse*[4]. You can find guided instructions here.

5. *Launch (start) Eclipse*. If you have not done this yet, select a workspace directory (you will be prompted). Note that special characters in the path of this directory are not recommended.

6. *Run a Java Hello World application*. On the welcoming screen, Eclipse provides tutorials. Select this and then select *Create a Hello World application* and complete the tutorial. If you need additional instructions on how to write a Hello World application in Java, refer to the source here.

Task-3: Installing and executing MARBEL

The purpose of this task is to install the MARBEL agent programming language for programming cognitive agents. This exercise helps you to set up MARBEL on your system. For this task to succeed, you must have been able to successfully complete Task 2. We assume you still have Eclipse open, if not, launch it again. Please follow these steps:

1. *Add a link to the MARBEL repository in Eclipse*:

   a. In the Eclipse menu, select *Help>Install New Software>Add*.

   b. Add https://goalapl.dev/marbel in the 'work with:' field and click the *Add* button.

   c. Type in `MARBEL` in the name field and click the *Add* button.

2. *Check the box* in front of the MARBEL Agent Programming Language item that appears.

3. Click *Next* (twice) and continue the installation. You will get a warning about trusted authorities. Check the box and press the *Trust Selected* button (twice). At the end of the installation, press the *Restart Now* button.

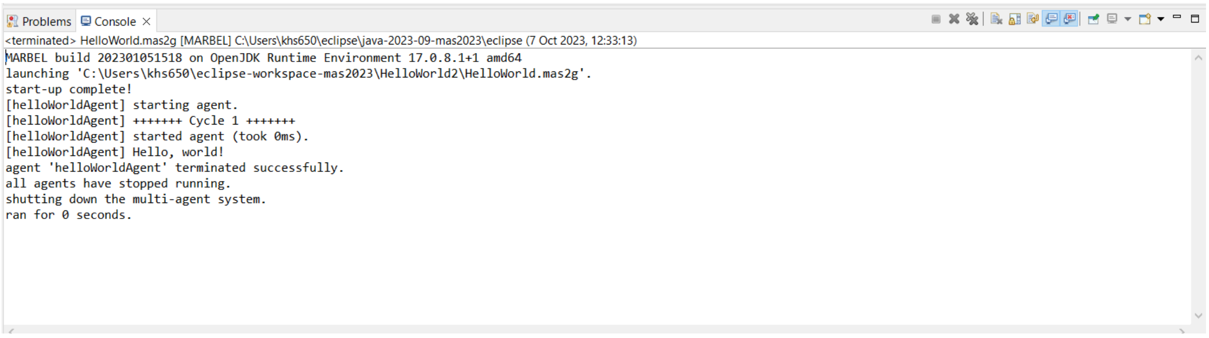
4. *Switch to the MARBEL perspective*. Use the *Window > Perspective > Open Perspective > Other... > MARBEL* menu to switch to the MARBEL perspective. Note that a shortcut is added in the top-right corner of Eclipse to switch to this perspective.

[2] You will need to create an Oracle account to be able to download Java.
[3] The release date of this Java 11 version is July 23, 2025.
[4] If you get an error (on Windows) that it cannot find / install NatTable extension for GlazedLists you can try installing an older version of Eclipse (e.g. Eclipse - IDE for Java Developers 2024-06).

Task-4: Executing an example MARBEL agent

The purpose of this task is to execute a Hello World agent in MARBEL to validate whether your system is set up correctly to execute MARBEL source code. We assume that you still have Eclipse with the MARBEL perspective selected open. If not, please launch it again.

1. *Open the example MARBEL Hello World agent*:
    a. Open the MARBEL Example Project Wizard by selecting *File > New > MARBEL Example Project* from the menu.
    b. In the pop-up window, select the *HelloWorld* project in the dropdown menu. Change the name to *HelloWorld2*[5] and press *Finish*. In the Eclipse *Script Explorer* window, there now should be a new project with the *HelloWorld2* name.
2. *Expand the new project*. Double-click on the project name in the *Script Explorer* window.
3. *Run the HelloWorld project*. Locate the file called `HelloWorld.mas2g`, right click on it, and select *Run As > 1 MARBEL*.
4. You should see similar output to that below in your Eclipse Console:



If you see `Hello, world!` in the console, then you have successfully executed your first MARBEL agent.

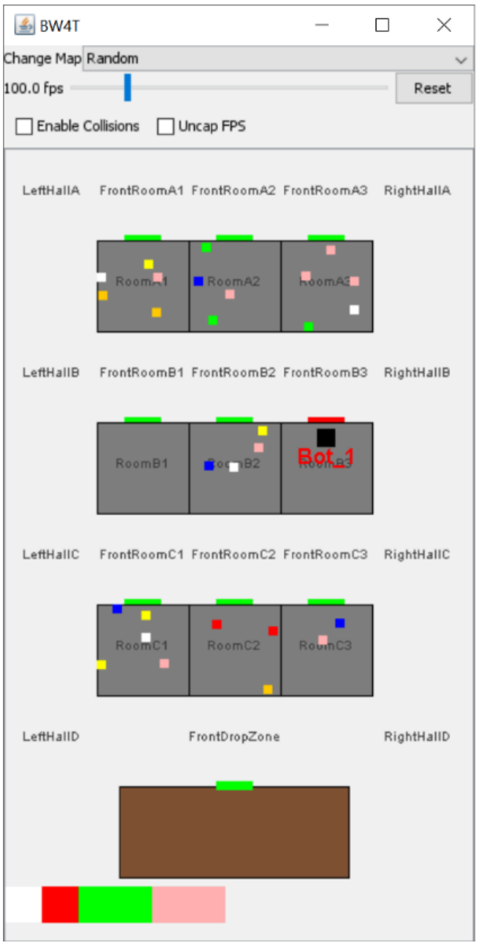
Task-5: Executing the BW4T project.

The purpose of this task is to get the simulation environment called Blocks World for Teams (BW4T) installed on your system. You need this environment for completing assignment 3 of the course. We assume that you still have Eclipse with the MARBEL perspective selected open.

1. *Open the BW4T project*. Open the MARBEL Example Project Wizard, select the *BW4T* project, and click *Finish*.
2. *Download the BW4T server*. Download the `bw4t-server-3.10.1.jar` file from here.
3. *Copy and paste the BW4T server into the MARBEL project*. Copy the file you just downloaded, right-click the BW4T project in the *Script Explorer*, and select *Paste*.
4. *Start the BW4T server*. Start the server either by double-clicking on the jar file in the Script Explorer, or by using the `java -jar` command in a terminal (go to the location of the jar

[5] We already created a Java Hello World project in Task 2 and Eclipse needs a different name to avoid a conflict.

file and execute, e.g., `java -jar bw4t-server-3.10.1.jar`). You should now see a GUI similar to the one below:



5. *Run the* `BW4T.mas2g` *file*. Before running the file, we suggest resizing the Eclipse window, so you can see both that window and the BW4T GUI. To run the `mas` file in Eclipse, right-click on the `BW4T.mas2g` file, and, this time, select the *Debug As* menu item, and click '1 MARBEL' (or, alternatively, use the Debug icon in the toolbar menu after selecting the file).
   In case of error, check the NOTE below.

6. *Switch to the MARBEL Debug perspective*. Use the *Window > Perspective > Open Perspective > Other... > MARBEL Debug* menu to switch to the MARBEL debug perspective.

7. You should see in the server GUI that a bot is trying to move (it is somewhat shaking) in the environment. If you can observe this, then your environment works!

THE END.

NOTE: (Only relevant for MAC users.) When executing `BW4T.mas2g`, if you encountered an error `MARBELRunFailedException` with cause: `bw4t-client-3.10.1.jar` does not exist, then your IDE cannot locate the `bw4t-client-3.10.1.jar` file. To fix this, you need to provide the full path to the jar file in the `mas2g` file (in line 1) as below:

Replace `use "bw4t-client-3.10.1.jar" as environment with` with:

```
use "<YOUR_PATH>/bw4t-client-3.10.1.jar" as environment with
```

where <YOUR_PATH> points to the right location, e.g., /jdoe/workspace/eclips/BW4T